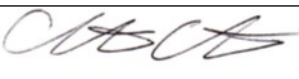


**Guideline Title: Rapid Sequence Induction (RSI) for
Endotracheal (ET) Intubation**

Guideline Number: GUID-3203-G008

Issue date: 6/1/2013	Replaces Dept. Policy:
Revision dates: 09/09/2014, 2/7/2017, 6/8/2021, 8/6/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Rapid sequence intubation (RSI) is the preferred approach to endotracheal intubation for any patient with a gag reflex or muscle tone. Special care should be taken to evaluate the indication for RSI and the potential for difficulties securing the airway. A bag-valve mask hooked to oxygen, a gum elastic bougie or endotracheal tube introducer, and a supraglottic airway device must always be kept easily accessible to the provider intubating the patient for rescue purposes. *Video laryngoscopy should be used for all endotracheal intubations unless it is not operationally feasible.*
- II. **DEFINITIONS:**
 - a. ACT- Air Care Team
 - b. BVM- Bag valve mask
 - c. ETT- Endotracheal tube
 - d. GCS – Glasgow Coma Scale
 - e. RSI- Rapid Sequence Intubation
 - f. SBP - systolic blood pressure
- III. **DEPARTMENT GUIDELINE:**
 - a. Patient Selection
 - i. The following patients shall be intubated using RSI:
 1. Actual or potential airway compromise due to depressed sensorium (GCS of 8 or less) or whose combativeness threatens the airway, spinal cord stability, or transport safety
 2. Patients who demonstrate a high probability of airway compromise during transport (i.e. smoke inhalation, severe facial trauma with bleeding)
 3. Patients who need ventilator assistance or airway protection
 - ii. The following patient should not be intubated using RSI:
 1. Patients that cannot be manually ventilated
 2. Any patient who is maintaining a patent airway or has spontaneous respirations of at least 12 breaths per minute
 - iii. The following patients *should not* be given Succinylcholine for intubation:
 1. Penetrating eye injuries or history of glaucoma

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2. Neurological disease (myasthenia gravis or multiple sclerosis)
3. Severe burn or trauma greater than 48 hours old
4. History of malignant hyperthermia
5. History of renal failure
- iv. Prior to RSI administration, careful consideration needs to be given to any patient where there is a concern for a difficult airway
- v. Induction agents may lower blood pressure. Ensure you are resuscitating the patient to maintain perfusion and oxygenation throughout the procedure.
- b. Procedure
 - i. Gather equipment and prepare patient per *GUID3203-PR020 - Endotracheal Intubation*
 - i. Pre-oxygenate with high flow oxygen by mask. If available, simultaneously oxygenate with nasal cannula. Do not manually ventilate the patient unless respiratory effort is ineffective, as this may cause gastric distention with vomiting and aspiration
 - ii. Ensure patient is connected to cardiac monitoring. Obtain BP, HR, and oxygen saturation
 - iii. Pre-medicate pediatric patients < 5 years of age who are not markedly tachycardic (> 160 bpm) with **Atropine 0.02 mg/kg IV**, maximum dose 0.4 mg, to prevent bradycardia associated with laryngoscopy.
 - iv. Push Dose Pressor (PDP, for Adult patients only): If the patient has a systolic blood pressure less than 90 mmHg after 250mL-500mL fluid bolus, administer **Phenylephrine 200 mcg IV push** prior to administering induction and paralytic agents.
 1. Phenylephrine is contraindicated if the heart rate is < 60 bpm
 2. May repeat every two minutes if the SBP remains less than 90 mmHg on repeat vital signs
 3. PDPs provide a transient improvement in pressure meant to be a bridge to safely perform RSI and continue resuscitation, *they may only be used for documented hypotension 5 minutes before to 5 minutes after the RSI procedure.*
 - a. For persistent hypotension more than 5 minutes after RSI procedure, treat per *GUID3203-M011 Hypotension* or *GUID3203-PR025A/B Emergency Blood Product Administration*
 4. PDPs may only be administered from approved prefilled syringes obtained from pharmacy – it is prohibited for any RN or Paramedic to mix their own PDP
 - v. RSI requires both an induction agent AND a paralytic agent be given in sequence as below. Do not administer paralytic agents for intubation to a patient that has not been given an induction medication first. Patients with a GCS of 3 or in cardiac arrest may be intubated without induction or paralytic medication.
 - vi. For induction of sedation, administer either:
 1. **Etomidate 0.3 mg/kg IV/IO** ****OR****
 2. **Ketamine 1-4.5 mg/kg (start at 2mg/kg for adults) IV/IO.** Consider when hypotension is present.

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- vii. Next, for muscle paralysis prior to intubation administer either:
 - 1. Adults: **Succinylcholine 1-1.5 mg/kg IV/IO**
Pediatrics: **Succinylcholine 2 mg/kg IV/IO** ****OR****
 - 2. **Rocuronium 0.6-1.2 mg/kg IV/IO**
 - 3. Administer over 30 seconds and wait until fasciculation stops
- viii. Consider removing cervical collar (if present) while providing in-line manual immobilization of the head and neck
- ix. Perform intubation per *GUID3203-PR020 - Endotracheal Intubation*
 - 1. If unsuccessful, supply O2 with BVM for 30-60 seconds and re-attempt intubation
 - 2. If patient begins to clench teeth or move, administer second dose of Succinylcholine prior to second intubation attempt
 - 3. Treat bradycardia during intubation by temporarily halting attempt and hyperventilating the patient with the BVM and 100% oxygen. If patient remains bradycardic, consider **Atropine 0.5 mg IV (adult) or 0.02 mg/kg IV (pediatric)** to a max single dose of 0.5 mg and a maximum total dose of 2 mg
 - 4. If endotracheal intubation fails after 2 attempts and you are unable to adequately ventilate the patient with a BVM, consider a supraglottic airway device See *GUID3203-PR016 - King LTS-D*. Otherwise, continue to ventilate with BVM.
 - 5. If the supraglottic airway device insertion is unsuccessful and the patient is still unable to be ventilated with a BVM, consider cricothyroidotomy using the H&H Emergency Cricothyroidotomy Kit. See *GUID3203-007 - Cricothyroidotomy-Surgical*.
- x. Once intubation is completed, inflate balloon (if present), confirm tube placement with a waveform ETCO2 detector.
- xi. Assess epigastric sounds and breath sounds. Secure tube.
- xii. All advanced airways (ETT, King LT, iGel, Laryngeal mask airway [LMA]) *must* be monitored with continuous waveform capnography during ACT transport.
 - 1. If no waveform is present and brief equipment troubleshooting does not result in a four phase capnographic waveform, REMOVE THE ADVANCED AIRWAY DEVICE and attempt reinsertion or ventilate patient with BVM
- xiii. Reapply cervical collar and complete spinal immobilization for transport
- xiv. Assure that all intubated patients have protective devices applied to wrists and/or ankles to protect patient and crew
- xv. For continued sedation during transport:
 - 1. **Midazolam 0.05-0.1 mg/kg IV/IO**, may repeat every 3-5 minutes PRN, maximum dose 5 mg. Maintain SBP > 90 mmHg
 - 2. If SBP < 90 mmHg, consider **Ketamine**
 - a. **Adults: 1-2 mg/kg slow IV/IO**. May repeat every 15 minutes PRN. Maximum total dose 200 mg. Monitor for tachydysrhythmias

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- xvi. Consider **Rocuronium 0.6-1.2 mg/kg IV/IO (adult) or 0.6 mg/kg IV/IO (peds)** for ETT/airway protection and prolonged paralysis during transport if patient is intubated and has adequate sedation. ****Do not give a paralytic drug if patient cannot be sedated****
- xvii. For pain control and comfort, manage per *GUID3203-G005 – Analgesia and Sedation Management*
- xviii. Provide post-intubation care per *GUID3203-PR020 - Endotracheal Intubation*
- xix. Monitor for complications including vomiting, cardiac dysrhythmias (bradycardia, PVCs, and V-fib), malignant hyperthermia, initial fasciculations, failed intubation, and aspiration of stomach contents

IV. DOCUMENTATION:

- a. Patient report/EMS Charts
- b. Narcotic Log/PSTrax

V. REFERENCES:

- a. Frakes, M.A., (2004). Clinical Features of Patients Intubated by a Flight Team. *Air Medical Journal* 23:3:37-39.
- b. Poste, J.C., et al, (2004). Air Medical Transport of Severely Head-Injured Patients Undergoing Paramedic Rapid Sequence Intubation. *Air Medical Journal* 23:4:-40.
- c. Smith, C. E., et al, (2002). Prehospital Tracheal Intubating Conditions during Rapid Sequence Intubation: Rocuronium versus Vecuronium. *Air Medical Journal* 21:1:26-32.
- d. Wang, H.E., et al (2001). Prehospital Rapid Sequence Intubation. *Prehospital Emergency Care* 5:40-48.